Correctness Scores Alone Are Unreliable Safety Certificates

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Abstract

Process reward models (PRMs) score the local correctness of reasoning steps, and correctness scores are natural candidates for deciding whether a step is safe to subject to a counterfactual rollback. We test whether they can *license* such safety decisions—not whether reasoning can be compressed in general. Our counterfactual protocol—control, deletion, position-and-length-matched *placebo* deletion with its own matched control, and semantic-preserving *paraphrase*—is applied to 600 rating-balanced PRM800K steps, with replications on two further generators, a within-checkpoint thinking-mode toggle, and 300 ProcessBench steps (over 23,000 regenerated continuations in total). Deletion gains concentrate almost entirely in steps that are both low-rated and semantically harmful ($+23.3$ pp; difference-in-differences $+25.4$ pp): erroneous steps act as *negative anchors* for no-thinking generation. An apparent generic "restart" benefit under a shared control disappears once each placebo cutoff is matched to its own control (1,513 of 1,514 returned), and paraphrasing the erroneous step performs like keeping it. Toggling the same generator into extended thinking makes the anchor vanish ($+21.8 \rightarrow -3.0$ pp). Turning scores into rollback policies, we budget the net accuracy change per rollback under a natural-prevalence reference weighting with nested train/calibrate/test selection: the resulting certificates are split-unstable, and their calibration is checkpoint-dependent—one trained PRM certifies in every split yet meets its budget on only 9 of 20 test splits (realizing -6.3 pp per rollback), while another transfers on most—so correctness scores alone cannot be trusted as safety certificates a priori. A single probe-and-rollback generation removes all detectable excess harm in a gold-verifier simulation. Correctness scores nominate rollback candidates; alone, they are unreliable safety certificates. We release the audited dataset, pipeline, and benchmark.

1 Introduction

Process reward models assign scores to individual steps of a reasoning trajectory and have become a standard tool for verifying and steering large language model (LLM) reasoning (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024). Work on reasoning compression motivates asking whether an intermediate step can be safely altered, but compression quality is not our estimand. Our narrower question is whether a PRM correctness score can safely trigger a counterfactual rollback. Reusing correctness scores as rollback signals embeds a silent assumption: that a step's local correctness tells us what happens to downstream reasoning if the step is *removed*. Before that reuse hardens into practice, we test the assumption directly.

However, evaluating rollback safety is complicated by a second issue: deleting a reasoning step does not only remove its content, but also changes the generation state. A deletion may appear beneficial because it removes harmful information, or simply because it provides the generator with a different opportunity to restart. Therefore, isolating semantic contribution requires counterfactual controls that separate step-specific effects from generic regeneration effects.

This assumption conflates distinct properties of a step: it can be locally correct yet contribute nothing later steps use; locally incorrect yet arrive after the trajectory is already doomed; or incorrect in a way that actively *anchors* the continuation to a bad path, so removing it rescues the run. Whether rollback is safe is a *counterfactual* property of the step–generator pair, not a syntactic property of the step in isolation. This also raises a broader question: whether negative anchors reflect an intrinsic property of erroneous reasoning steps, or an interaction between step errors and the generator's ability to recover during inference. We therefore examine whether additional reasoning computation changes the observed rollback behavior.

Main question and answer. Can PRM scores safely decide which reasoning steps to remove? We find that correctness scores alone cannot: they detect local wrongness, but do not determine semantic contribution or counterfactual removability. Safe rollback therefore requires counterfactual validation rather than a correctness threshold alone. Importantly, a retrospective deletion effect does not automatically translate into a deployable policy: a system must know whether a score can certify safety before intervention under realistic risk constraints.

We study this gap directly. Starting from PRM800K (Lightman et al., 2024), we sample 600 target steps balanced across human ratings ($-1/0/+1$), annotate each step's semantic contribution (Essential / Redundant / Harmful), and measure what actually happens when a fixed generator (Qwen3-8B; Qwen Team, 2025) continues the trajectory with and without the step, four runs per condition, 2,400 paired outcomes. Because deleting *any* step also forces the generator to re-plan—which can itself change outcomes—we additionally run a position-and-length-matched *placebo* deletion for 511 of the 600 steps (1,514 runs), decomposing raw deletion effects into target-specific and generic-restart components. An LLM judge scores final answers, audited against 200 human adjudications (headline sensitivity ≤ 0.13 pp).

Three findings answer this question. **First—PRM correctness is not removability.** Correctness and contribution are correlated but far from interchangeable (Cramér's $V = 0.53$; 34.5% of steps off the expected diagonal), and deletion gains concentrate overwhelmingly in steps that are both low-rated and Harmful ($+23.3$ pp): the rating-0, rating-$+1$, Essential, and Redundant groups are statistically indistinguishable from zero, and the small off-diagonal cells are inconclusive. The pattern replicates on a second no-thinking generator (phi-4: anchor effect $+33.7$ pp) and on external ProcessBench trajectories (a twelve-model benchmark; six source models in our sample); toggling the *same* Qwen3-8B checkpoint into thinking mode collapses the anchor effect ($+21.8 \rightarrow -3.0$ pp), as does a long-CoT distill—extended thinking itself neutralizes bad prefixes. **Second—deletion gains come from negative anchors.** A single-control placebo suggests a $+10.0$ pp generic "restart" component, but that contrast is confounded: half the matched cutoffs fall before the target and silently remove it. Giving each placebo cutoff its *own* control (1,513 of 1,514 returned) shows matched innocuous deletions are inert in every group of this cohort, and the anchor effect is fully target-specific ($+25.4$ pp, difference-in-differences); the paraphrase control shows it travels with meaning: rewording the erroneous step performs like keeping it (a *negative anchor*). **Third—rollback needs counterfactual validation.** We budget the *certified net accuracy change per rollback* (a one-sided 95% cluster bound), which repairs a subtle flaw in transition-rate "harm": independent regeneration makes a null rollback on a stochastic step look 25% harmful. Under deployment weighting with nested train/calibrate/test selection over 20 splits, correctness-based certificates are unstable (even random certifies in 9/20 splits, meeting the budget in only 2), and transfer is checkpoint-dependent: one trained PRM certifies in every split yet meets its 1 pp budget in only 9 of 20 on test (-6.3 pp per rollback realized); another transfers in 16 of 19—and correctness quality does not predict which (three trained checkpoints). The rating-$+1$ mass that a correctness-based rollback policy would most often label as safe (92% of natural traffic) never certifies below a 5 pp budget, and no deployable signal certifies it broadly—the safest covers only the error tier ($+20.8$ pp at 13% coverage), and the one signal that covers more reads the deletion outcome itself. The information gap, not the intervention, is the bottleneck—and in a gold-verifier simulation, a single probe-and-rollback generation removes all detectable excess harm while raising the net gain to $+11.5$ pp per candidate.

Our contributions follow this main line: (i) we separate PRM correctness, semantic contribution, and retrospective removability and show that correctness does not imply removability; (ii) we provide a counterfactual protocol that isolates negative-anchor mechanisms by separating semantic effects from regeneration effects through matched placebos, own controls, and semantic-preserving paraphrases; and (iii) we turn the construct gap into a rollback risk–coverage benchmark, showing when correctness signals fail to certify safety and how a cheap probe-and-rollback check repairs the risk. We release all data, code, and audits (Appendix C).

Label convention. Unless otherwise stated, all main results use the frozen AI-assisted semantic labels (Essential 165 / Redundant 201 / Harmful

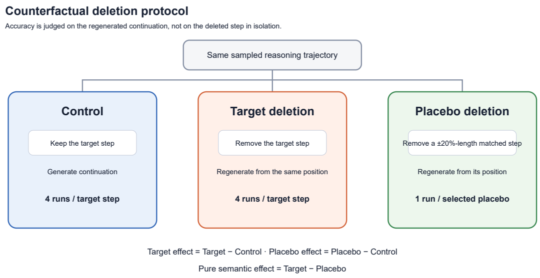

Figure 1: Study logic and counterfactual protocol. PRM rating (local correctness) is correlated with, but not equivalent to, semantic contribution; counterfactual removability is a further layer measured by regenerated Control, Target-deletion, and matched Placebo continuations. A correctness rating may nominate a candidate but is not itself a rollback-safety certificate.

234). Human-calibrated labels (198 / 199 / 203) are used only for explicitly identified robustness or diagnostic analyses and do not replace the primary label set.

The conceptual chain is *PRM rating* $\longrightarrow$ (correlated but non-equivalent) *semantic contribution* $\longrightarrow$ (not determined by either) *counterfactual removability*. Figure 1 puts this distinction and the intervention safety test before the detailed results.

2 Related Work

Process supervision and PRMs. Step-level supervision improves credit assignment over outcome-only rewards (Uesato et al., 2022; Lightman et al., 2024; Cobbe et al., 2021). Math-Shepherd derives step labels from rollout success probabilities (Wang et al., 2024); Setlur et al. (2025) argue for rewarding *progress* (prospective advantage) rather than local correctness; generative PRMs verify steps with explicit reasoning (Khalifa et al., 2025). Benchmarks probe what PRMs detect: ProcessBench locates first errors (Zheng et al., 2025), and PRMBench separates fine-grained error dimensions including redundancy (Song et al., 2025). None measures what happens when a step is *removed*—the counterfactual layer our protocol supplies.

Reasoning interventions and rollback safety. Token- and step-pruning methods study efficiency-oriented alterations to chains of thought using attention-based redundancy (Choi et al., 2025), step entropy (Li et al., 2025), or likelihood-preserving greedy deletion (Singh and Hakkani-Tür, 2026). We do not evaluate compression efficiency or claim that reasoning cannot be compressed. Instead, we test the narrower safety question raised when PRM correctness is reused as a rollback trigger: does local correctness certify the downstream safety of removing a step? These works primarily preserve the answer of an already-correct trajectory; we additionally quantify the *repair* regime and the safety of the intervention itself.

Deletion as intervention. Ablation-style causal analyses of reasoning typically mask content at the token level or measure answer-probability shifts. Our design instead runs full *regenerated* continuations—the deployment-relevant intervention—and controls the regeneration with matched placebos that carry their *own* controls, which proves essential: a shared-control placebo manufactures an apparent $+10$ pp "restart" component that disappears, and understates the semantic effect, once each cutoff is read against its own control (Section 6).

3 Constructs and Estimands

We distinguish four properties of a step s_t in trajectory h: (1) *local correctness*, the PRM800K rating in $\{-1, 0, 1\}$; (2) *semantic contribution*, whether the content of s_t is needed by (Essential), irrelevant to (Redundant), or misleading for (Harmful) downstream reasoning; (3) *prospective advantage*, the change in success probability from *taking* the step (Setlur et al., 2025); (4) *retrospective removability*, the change in success probability from *deleting* the step after the fact and regenerating. This paper measures (4) and its relation to (1) and (2).

For step i, let $Y_{ij}^c, Y_{ij}^t \in \{0, 1\}$ be final-answer correctness under **C**ontrol (step kept) and **T**arget deletion, with $K{=}4$ independent runs per condition, $\bar{Y}$ denoting means over runs and p_i^c, p_i^t the underlying success probabilities; matched steps (511/600) additionally receive one to four **P**lacebo runs Y_{ij}^p (1,514 in total), applying the same truncation operator at a different, length-matched step p, and every placebo cutoff receives an *own control* C_p (prefix kept through p; 1,513 of 1,514 scheduled runs returned). Pre-specified estimands (fixed in the released analysis plan before the extension cohorts ran; we did not use an external registry):

$$\Delta_i^{\text{tgt}} = \bar{Y}_i^t - \bar{Y}_i^c, \qquad \Delta_i^{\text{plc}} = \bar{Y}_i^p - \bar{Y}_i^c,$$
$$\Delta_i^{\text{sem}} = \Delta_i^{\text{tgt}} - \Delta_i^{\text{plc}}, \tag{1}$$

plus the own-control difference-in-differences $\Delta_{\text{DiD},i}^{\text{sem}} = \Delta_i^{\text{tgt}} - (\bar{Y}_i^p - \bar{Y}_i^{C_p})$, adopted after the shared-control confound in Δ^{plc} was discovered (Section 6; the pre-specified contrast is retained as *legacy* in Table 14, appendix), plus run-level *discordance* events *danger* ($Y^c{=}1$, $Y^t{=}0$ on independent draws) and *benefit* ($Y^c{=}0$, $Y^t{=}1$). Runs are independent fresh-session samples with no shared randomness across conditions; pairing control run j with deletion run j is a fixed bookkeeping convention whose expected discordance rate equals $p_i^c(1{-}p_i^t)$ under any pairing, so danger/benefit rates estimate those products—independent-sample discordance, never within-seed counterfactual flips (Section 9 builds the causal safety metric). Discordance is reported either as a joint rate (share of all pairs) or conditioned on the control draw's outcome; we state which at each use. All confidence intervals are 95% percentile intervals from 5,000-replicate cluster bootstraps over *target steps*; re-clustering by problem leaves headline intervals essentially unchanged (Table 14, appendix). Fixed seeds are released.

4 Data and Intervention Protocol

Step cohort. From PRM800K phase-2 test trajectories (Lightman et al., 2024) over MATH problems (Hendrycks et al., 2021) we sample 600 target steps, 200 per rating, balanced over early/middle/late positions ($201/201/198$). **Unless otherwise stated, all main results use the frozen AI-assisted semantic labels; human-calibrated labels are only used for explicitly identified robustness or diagnostic validation.** Semantic contribution was annotated twice, with both passes completed before any deletion outcome existed: an initial AI-assisted pass (165 Essential / 201 Redundant / 234 Harmful) and a human-calibrated re-labeling pass (198 / 199 / 203); the passes disagree on 164/600 steps (27.3%), an explicit reliability bound; a blinded four-annotator pilot provides a human-only robustness check (Table 5; full agreement details in Appendix B). The frozen statistical pipeline behind Tables 3–6 uses the *initial* pass as its primary label set ($-1{\times}$Harmful anchor: 178 steps); the calibrated pass (anchor: 160 steps, 154 shared) is confined to the correctness-vs-contribution diagnostic, extension-cohort sampling metadata, and explicitly labeled robustness re-estimations (Table 14, appendix; the headline survives either pass). Under the calibrated labels, rating and type are associated (Cramér's $V = 0.528$) with 65.5% of steps on the expected diagonal—correlated, but with a third of the cohort off-diagonal (Figure 3, appendix).

Position	Steps	Target–Control (pp)	95% CI (pp)
Early	201	$+5.0$	$[-1.1,\ +11.2]$
Middle	201	$+9.5$	$[+3.3,\ +15.6]$
Late	198	$+7.2$	$[+1.8,\ +12.6]$

Table 1: Position-binned target-deletion effects in the primary cohort. Estimates and 95% percentile cluster-bootstrap intervals are from the frozen analysis artifact; the bins were balanced by design. The effect is not confined to early positions.

Human-only label robustness. The pilot was blinded to PRM ratings, prior labels, and deletion outcomes. Two independent annotators labeled each of 120 steps; pooled agreement was 61.7% ($\kappa = 0.40$), so this is a sensitivity analysis rather than a replacement annotation of the full cohort. Among the 74 exact-consensus items, the 15 consensus-Harmful steps show a $+25.0$ pp target-deletion effect and a $+22.9$ pp legacy placebo-corrected effect; the corresponding effects are small for consensus-Essential steps ($+4.2$ / $+2.6$ pp) and reverse for consensus-Redundant steps (-22.1 / -5.8 pp). Thus the Harmful-anchor pattern is reproduced by labels assigned without access to outcomes. We report the human-consensus/deletion-gain relationship in the main results (Table 5); the small and only moderately reliable pilot does not justify treating the semantic taxonomy as ground truth.

Position robustness. The primary cohort was balanced across early, middle, and late target positions, allowing a direct check that the aggregate effect is not an early-step artifact. The frozen step-cluster bootstrap gives positive point estimates in all three bins, with overlapping intervals (Table 1).

Anchor-by-position interaction. To rule out that negative anchors are an artifact of their location in the chain, we reuse the same early/middle/late bins and split each bin into the primary $-1{\times}$Harmful anchor and all other steps. Anchor effects are positive in every bin (early $+16.1$ pp, middle $+24.6$ pp, late $+28.8$ pp), while the pairwise bootstrap CIs for the Anchor–Other contrast differences all include zero (Table 10). The negative-anchor effect is therefore not confined to a particular chain position.

Random-label negative control. To test whether the Harmful result is merely a label-frequency artifact, we permuted each semantic-label vector 5,000

Statistic	Observed	Random-label 95% interval
Raw $-1\times$Harmful effect (pp)	$+23.3$	$[+14.0,\ +30.3]$
Within--1 Harmful contrast (pp)	$+12.0$	$[-12.9,\ +13.5]$
Calibrated candidate coverage	26.7%	$[9.5\%,\ 13.0\%]$
Calibrated candidate effect (pp)	$+20.9$	$[+13.0,\ +31.1]$

Table 2: Random semantic-label negative control. Each permutation preserves the initial margins (165/201/234) or calibrated margins (198/199/203), as appropriate, and uses the same 600 frozen outcomes. The within-rating contrast is the mean effect for rating--1 Harmful steps minus rating--1 non-Harmful steps; candidate coverage is the share selected by rating-$-1\times$Harmful. Intervals are the 2.5th–97.5th percentiles over 5,000 permutations (seed 20260729).

times while preserving its observed margins, leaving ratings and all frozen outcomes fixed. The raw rating-$-1\times$Harmful effect remains positive under this null because the rating--1 effect is deliberately retained; the sharper diagnostic is the within-rating contrast between randomized Harmful and non-Harmful steps (Table 2). This contrast is centered at zero under the permutation null. The observed label–rating intersection is also much more concentrated than random labels (26.7% versus a 9.5–13.0% null interval for the calibrated policy candidates). The candidate-effect interval overlaps the null because rating--1 itself carries much of the raw effect, so this ablation supports semantic-label enrichment and guards against frequency-only interpretations; it is not a claim that the label alone identifies safe rollbacks.

Interventions. Both interventions are prefix truncations of the same trajectory: the Control prompt contains the problem plus steps $1{:}t$, the Target-deletion prompt contains steps $1{:}t-1$—operationally a *rollback* of the most recent visible step, not an interior splice—and Qwen3-8B (temperature 0.7, top-p 0.8) regenerates the continuation, four fresh runs per condition. The Placebo condition applies the identical truncation at a different step p of the same trajectory (before or after t; 51%/49% in practice) whose token length is within $\pm 20\%$ of the target's, choosing the closest available position; 511/600 steps admit such a match (1,514 placebo runs), and the 89 unmatched steps differ from matched ones only modestly (largest imbalance: position SMD 0.298). The placebo-cutoff steps are benign: none carries rating -1 (Table 15, appendix).

Outcome evaluation and audit. An LLM judge (Qwen3-8B, temperature 0; prompts released) marks final-answer correctness against ground truth. We audited 200 stratified outputs with human adjudication: agreement 93.0%, maximum per-condition bias 2.3 pp—below the pre-set gates (90%/5 pp; detail in Table 8, appendix). Substituting all 200 adjudicated labels into the frozen tables moves the overall target effect by -0.08 pp and the placebo-corrected effect by -0.05 pp—observed sensitivity on the audited subset, not a bound on the remainder; no conclusion moves under substitution (Table 8). Because the judge shares a checkpoint with the primary generator, a deterministic PRM800K-style normalization+SymPy grader independently regrades all 6,314 primary runs: 93.0% agreement, headline effects within 0.9 pp (Appendix A).

Replication and mechanism cohorts. Four extensions reuse the identical prompt protocol. (i) *Cross-generator*: 300 steps (100 rating $-1\times$Harmful anchors, 100 rating-0, 100 rating-$+1$; 103 stably-correct; positions balanced) rerun with **phi-4** (14B), 3 runs per condition, reusing the frozen matched-placebo indices; the identical 2,691-task list is additionally rerun with **DeepSeek-R1-Distill-Qwen-14B** (long-CoT, served locally; sampling and judge unchanged) and—holding the checkpoint fixed—with **Qwen3-8B's own thinking mode enabled** (same provider, prompts, sampling; 3,582 runs incl. 891 own-controls).[1] (ii) *Semantic-preserving paraphrase*: for a 240-step strong-control subset (80 anchors / 80 stably-correct / 80 fragile-neutral, rating $0/+1$), an independent model (Gemini 2.5 Flash) paraphrases the target step preserving its exact meaning *including errors*; 221/240 paraphrases passed automatic validity checks (7.9% excluded) and each ran 4 continuations, alongside 4 fresh matched-placebo runs. Condition codes: C0 = frozen control, C1 = frozen target deletion, C2 = fresh matched placebo, C3 = fresh paraphrase. (iii) *Deployed PRM scoring*: five PRMs score all 600 steps—three trained checkpoints (Qwen2.5-Math-PRM-7B; Math-Shepherd; RLHFlow-Llama3.1-8B) and two prompted LLM-judge PRMs (Gemini 2.5 Flash, phi-4). (iv) *External validation*: 300 ProcessBench steps (Zheng et al., 2025) spanning GSM8K, MATH, OlympiadBench, and Omni-MATH—a benchmark whose

[1] R1-Distill was unavailable via the API provider at freeze time; phi-4 matched the 7–14B no-thinking budget (all 2,691 continuations parsed). The local long-CoT rerun parsed 99.0%; its 28 unparseable continuations, balanced across conditions, score as incorrect.

Group	Steps	Δ^{tgt} (pp)	95% CI
Overall	600	$+7.21$	$[+3.88, +10.58]$
rating $= -1$	200	$+22.00$	$[+15.41, +28.42]$
rating $= 0$	200	$+0.25$	$[-5.41, +5.90]$
rating $= +1$	200	-0.63	$[-5.54, +4.21]$
Essential	165	-1.21	$[-7.24, +4.78]$
Redundant	201	$+1.12$	$[-3.87, +6.06]$
Harmful	234	$+18.38$	$[+12.28, +24.44]$
$-1\times$Harmful	178	$+23.31$	$[+16.57, +30.24]$

Table 3: Target-deletion effects on final-answer accuracy (600 steps, 4,800 runs in 2,400 pairs, cluster bootstrap over steps). **All rows in this primary table use the frozen AI-assisted initial-pass semantic labels**; the human-calibrated anchor estimate (+20.9 pp) is a robustness result reported in Table 14 (appendix).

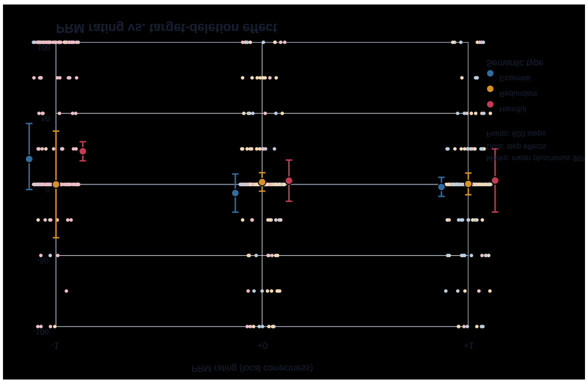

Figure 2: PRM rating versus target-deletion effect on the 600-step primary cohort. Each point is one step; colors show the frozen AI-assisted semantic label. Large positive effects concentrate in rating--1 Harmful steps, whereas rating-$0/+1$ groups cluster near zero. Larger markers show within-cell means with 95% step-bootstrap intervals.

trajectories come from twelve source models; six appear in our sample (Section 7).

5 Full-Cohort Deletion Results

Semantic-label ablation. To ask what semantic information adds beyond a PRM threshold, we condition on the 200 steps already assigned rating -1. The PRM-only candidate pool has a $+22.0$ pp deletion effect; adding the Harmful label narrows this pool to 178 steps (89% coverage) with $+23.3$ pp, while the 22 rating--1 non-Harmful steps show $+11.4$ pp (Table 4). The Harmful–non-Harmful mean contrast is $+12.0$ pp, but the incremental gain over the full PRM--1 pool is only $+1.3$ pp because Harmful steps already comprise 89% of the pool. The empirical positive-effect share rises only from 32.5% (65/200) to 33.7% (60/178), while the non-

Candidate set within rating -1	n	Coverage	Δ^{tgt} (pp)	Positive-effect share
PRM-only: all rating -1	200	100%	$+22.0$ $[+15.4, +28.4]$	32.5% (65/200)
PRM + Harmful	178	89%	$+23.3$ $[+16.6, +30.2]$	33.7% (60/178)
PRM + non-Harmful	22	11%	$+11.4$ $[-9.1, +31.8]$	22.7% (5/22)

Table 4: Two-stage selection within the PRM error pool. All rows use the same 200 rating--1 steps and the frozen primary AI-assisted semantic labels. Coverage is relative to that pool. Positive-effect share is the fraction of steps with observed $\Delta_i^{\text{tgt}} > 0$ across four independent runs; it is a descriptive enrichment measure, not a causal precision estimate. CIs are 5,000-replicate step-level cluster bootstraps (seed 20260802). Harmful steps have a larger mean effect than non-Harmful steps, but the incremental gain over the PRM-only pool is modest because Harmful steps comprise 89% of that pool.

Harmful subset is 22.7% (5/22). Thus this experiment does not show that Harmful is stronger than the PRM; it shows that semantic information provides modest within-rating enrichment for ranking which already-identified errors are more promising to delete. Because each step has only four independent runs, the positive-effect share is a descriptive enrichment measure, not a noise-free causal precision estimate.

Length-adjusted robustness. Because longer steps may be more disruptive to delete, we fit a problem-clustered OLS model over all 600 steps with rating indicators, initial semantic-type indicators, position indicators, and $\log(1+$ target tokens). The Harmful coefficient is $+7.1$ pp (95% CI $[-4.1, +18.4]$) and the log-length coefficient is only $+1.2$ pp per log unit (95% CI $[-4.9, +7.5]$); neither explains the anchor pattern after adjustment. Full coefficients are in Table 11 (appendix).

Deleting target steps helps on average ($+7.21$ pp, Table 3), but the average is misleading: the entire reliable gain sits in the rating $= -1$ column and the Harmful row, peaking at $+23.31$ pp for their intersection (the off-diagonal cells are small-n and inconclusive: the 22 rating--1 non-Harmful steps show $+11.4$ pp, CI $[-9.1, +31.8]$; the 56 Harmful non-(-1) steps $+2.7$ pp). Correct steps (rating $+1$) and Essential steps show near-zero *average* effects—yet not because deletion is safe there. Danger discordance is substantial everywhere: among paired draws whose control sample is correct, the deletion sample is wrong 15.1% of the time (a conditional rate—the joint rate is 8.5%, Table 16, appendix; neither is an individual causal flip, Section 3), and even within rating $= -1$ the conditional rate is 14.9%. The near-zero group means

Human consensus label	n	Rating -1 share	Deletion gain (pp)
Essential	42	21.4%	$+4.2$
Redundant	17	11.8%	-22.1
Harmful	15	93.3%	$+25.0$

Table 5: Human consensus versus target-deletion gain. The 120-step blinded pilot yielded 74 exact-consensus items; effects are averages over the frozen run-level outcomes for those consensus items. The corresponding legacy shared-control placebo-corrected effects are $+2.6$ pp (Essential), -5.8 pp (Redundant), and $+22.9$ pp (Harmful). Because the pilot has only 15 consensus-Harmful items and pooled agreement $\kappa = 0.40$, this is human-only robustness evidence, not a full-cohort human ground truth.

arise from offsetting discordance in both directions, not from inertness.

Human consensus and deletion gain. The outcome-blinded human pilot provides a direct semantic validity check on the intervention pattern. Among exact-consensus labels, consensus-Harmful steps show the largest deletion gain ($+25.0$ pp), consensus-Essential steps are close to zero ($+4.2$ pp), and consensus-Redundant steps show a negative average effect (-22.1 pp). The ordering is consistent with the proposed semantic interpretation, but it is a sensitivity result rather than a population-level correlation: only 74 of the 120 pilot items reached exact consensus, including 15 consensus-Harmful steps.

Trajectory state dominates raw effects. Stratifying by control stability makes the state-dependence explicit (Figure 4, appendix): steps whose four control runs are all wrong gain $+33.4$ pp from deletion (nothing to lose), while steps whose control runs are all correct *lose* 12.1 pp (CI $[-15.5, -8.9]$). These strata condition on the same control draws that enter Δ^{tgt}, so regression to the mean pushes the extreme strata outward; we read them as descriptive bounds on state-dependence, not causal subgroup effects—and a deployment policy never observes these counterfactual states, which is why raw deletion gains overstate what a policy can capture (Section 9).

Step-level stability. With four paired runs per step, 377/600 steps never change outcome; among low-rated Harmful steps, 29% are strongly beneficial to delete ($\geq$ 3-run gain) vs. 11% elsewhere—enrichment, not a guarantee.

Cross-generator replication. The pattern is not a Qwen3-8B artifact. Rerunning 300 steps with phi-4 (3 runs per condition) reproduces every qualitative conclusion: overall target effect $+12.4$ pp, anchor-group target effect $+33.7$ pp with own-controlled placebos inert (-1.3 pp) and a DiD semantic component of $+34.9$ pp (CI $[+25.8, +44.7]$), and rating-0/rating-1 target effects indistinguishable from zero ($+1.7$ and $+2.0$ pp). Effect *magnitudes* are generator-dependent, and step-level effect signs agree for 65% of the 80 steps with nonzero effects under both generators—well above chance but far from deterministic.

Long-CoT replication. The same 2,691 tasks under DeepSeek-R1-Distill-Qwen-14B—which thinks before answering—change the picture qualitatively. Anchor-step control accuracy rises to 77.7% (the model recovers from erroneous prefixes unaided), the overall deletion effect shrinks to $+3.4$ pp, and the anchor-group effect of $+7.0$ pp is not about the anchor at all: deleting a matched *innocuous* step helps just as much ($+8.0$ pp against its own control), and the DiD semantic component is -1.0 pp (CI $[-7.8, +6.3]$). Only 82/300 steps shift outcome at all: negative anchoring—and with it most of deletion's value—vanishes here, leaving only a small content-agnostic truncation benefit.

Thinking-mode contrast. The R1 replication changes model, scale, and training data at once; to isolate the mode, we rerun the same tasks on the *same* Qwen3-8B checkpoint with thinking enabled (identical provider, prompts, sampling; thinking activates on every task, mean 2,514 reasoning tokens; 12.2% of runs yield no parseable answer and are scored incorrect, balanced across conditions). On the anchor cell, whose no-thinking control accuracy is 26.8% with a $+21.8$ pp deletion effect, thinking lifts control accuracy to 83.3% and the deletion effect drops to -3.0 pp (CI $[-8.0, +2.3]$; DiD semantic -3.5 pp, n.s.); only 77/300 steps shift outcome at all, and step-level effect signs agree with the no-thinking runs at a rate consistent with chance (56%, n=32). The collapse of negative anchoring is attributable to extended thinking itself, not to the R1 checkpoint.

6 Mechanism: Anchors, not Restarts

Does deletion help because the *content* of the step was harmful, or because *any* deletion lets the gen-

Group	Δ^{tgt}	$\Delta^{\mathrm{plc}}_{\mathrm{vs\,C}}$	$\Delta^{\mathrm{plc}}_{\mathrm{own}}$	$\Delta^{\mathrm{sem}}_{\mathrm{DiD}}$ (CI)
Overall (511)	$+7.97$	-0.57	$+0.46$	$+7.52$ $[+2.8, +12.4]$
rating $= -1$ (169)	$+22.78$	$+9.47$	-0.69	$+23.47$ $[+14.4, +32.5]$
rating $= 0$ (176)	$+0.99$	-5.63	$+2.84$	-1.85 $[-9.7, +6.0]$
rating $= +1$ (166)	$+0.30$	-5.42	-0.90	$+1.20$ $[-5.9, +8.4]$
Harmful (201)	$+18.41$	$+3.57$	-1.74	$+20.15$ $[+12.0, +28.4]$
$-1\times$Harmful (151)	$+23.84$	$+9.99$	-1.60	$+25.44$ $[+16.1, +34.9]$

Table 6: Placebo decomposition (pp; initial-pass labels). $\Delta^{\mathrm{plc}}_{\mathrm{vs\,C}}$: placebo vs. the *target's* control (frozen contrast); $\Delta^{\mathrm{plc}}_{\mathrm{own}}$: vs. its *own* control C_p (1,513 new runs); $\Delta^{\mathrm{sem}}_{\mathrm{DiD}} = \Delta^{\mathrm{tgt}} - \Delta^{\mathrm{plc}}_{\mathrm{own}}$. Matched innocuous deletions are inert in every group of this cohort.

erator restart and resample? Table 6 answers with two placebo contrasts. Against the target's control, placebo truncation inside the anchor group looks strongly positive ($+9.99$ pp; rating -1 overall $+9.47$)—the reading a single-control design would report as a "restart benefit". But this contrast confounds removing step p with moving the cutoff: 51% of matched cutoffs fall *before* the target, so those placebo prefixes silently drop the erroneous step too. Splitting by side is diagnostic: cutoffs before the target yield $+21.1$ pp against the shared control (they remove the anchor), cutoffs after it yield -6.8 pp (they keep it). Giving each placebo cutoff its *own* control—new C_p runs that keep step p and truncate after it—removes the confound: $\Delta^{\mathrm{plc}}_{\mathrm{own}}$ is null in every group (-1.6 pp on anchors, $+0.5$ pp overall), and the difference-in-differences semantic effect *rises* to $+25.4$ pp (CI $[+16.1, +34.9]$) on anchors—essentially the entire raw effect: what matters is whether the *anchor* leaves the prefix. (An identical-prompt subset bounds generation-batch drift at $+3.1$ pp, within noise; Table 18, appendix.)

Two methodological corollaries: placebo deletions need their own controls—matching position and length is not enough (the single-control contrast *understated* the anchor's semantic component by $\sim$12 pp and manufactured a spurious restart component)—and the rating-$0/+1$ "semantic" effects of the single-control analysis ($+5.7$ to $+6.6$ pp) vanish under DiD (-1.9 to $+1.2$ pp, n.s.): target-specific removability lives *only* in the anchor group.

Paraphrase triangulation. The placebo isolates *that* the target matters; a semantic-preserving paraphrase isolates *why* (Table 9, appendix). In the anchor group, replacing the erroneous step with an independent-model rewording that preserves its (wrong) meaning leaves accuracy at control level ($+1.7$ pp, CI $[-6.2, +9.9]$), while deleting it gains $+19.2$ pp over the paraphrase (CI $[+8.2, +30.1]$): the harm travels with the *semantics*, not the surface form. The fragile-neutral group shows the mirror image: a large raw gain ($+24.1$ pp) that placebo—and even paraphrase—matches (C1$-$C2 $= 0.0$ pp): nothing target-specific, just conditioning on fragile control runs. The stably-correct group's vs-C0 contrasts inherit the same conditioning bias in the opposite direction (the group is selected on its C0 draws, so regression to the mean pushes C1$-$C0 and C3$-$C0 negative); its unconditioned contrasts show deletion $\approx$ paraphrase (C1$-$C3 $= -2.7$ pp, n.s.) and an inert operator (C2$-C_p = -0.8$ pp), so we do not read the -8.3 pp paraphrase row causally. A blinded 140-item human fidelity audit backs the C3 premise (97.1% faithful or minimally deviating; zero of 46 audited anchor-group paraphrases corrected the original error), and the 19 excluded paraphrases resemble the 221 retained (Table 20, appendix).

Eight fully human-verified cases (78 audited continuations) ship with the released artifacts.

7 External Validation on ProcessBench

Does any of this survive outside PRM800K trajectories and outside human step ratings? We sample 300 ProcessBench steps (Zheng et al., 2025)—152 human-annotated *first-error* steps and 148 *locally-correct* steps, balanced across GSM8K, MATH, OlympiadBench, and Omni-MATH (75 each), drawn from a benchmark whose trajectories span twelve source models (six appear in our sample)—and run the identical protocol (3 runs per condition; 204/300 steps admit a length-matched placebo; gold answers matched for 99.5% of records).

The core dissociation transfers. Deleting first-error steps yields $+24.8$ pp (CI $[+16.9, +32.7]$); on the placebo-matched subset, the difference-in-differences semantic component is $+35.4$ pp (CI $[+22.8, +48.9]$)—though here, unlike every no-thinking PRM800K cohort, the own-controlled placebo is itself significantly *negative* (-8.4 pp, CI $[-15.9, -1.4]$), so part of the DiD reflects a truncation penalty and the raw $+24.8$ pp is the placebo-free reading. Either way, erroneous steps in other models' reasoning are negative anchors too. Deleting locally-correct steps yields -2.0 pp raw (CI $[-8.3, +4.1]$) and a -7.1 pp (n.s.) semantic component: not usefully removable. One boundary emerges: the DiD semantic component is significant on GSM8K (+37.3 pp) and MATH (+21.1 pp) but attenuates on OlympiadBench (+6.0 pp, CI crosses zero) and vanishes on Omni-MATH (−0.6 pp)—and own-controlled placebos show no restart benefit there either (+6.3 pp, n.s.): on problems near the generator's competence ceiling, nothing target-specific survives. Negative anchors are a phenomenon of *recoverable* trajectories.

8 Do Correctness Signals Certify Rollback Safety?

If PRM-style signals suffice to trigger safe rollback, they should predict the two discordance events a policy cares about: *danger* (a correct control draw paired with a wrong deletion draw) and *benefit* (the reverse), both conditioned on the control draw's outcome. We evaluate step-level signals with problem-grouped 5-fold cross-validation, L2-regularized logistic models, and cluster-bootstrap CIs (Table 12, appendix).

The asymmetry matters. Signals rank *benefit* discordance moderately well: low rating and Harmful type genuinely enrich for rescuable runs. But for *danger* discordance—the safety-relevant direction—the rating alone is anti-informative (AUROC 0.422): among runs whose control draw is correct, low-rated steps are *not* safer to delete. Only features that peek at trajectory state (control stability; effectively extra compute) move danger prediction appreciably. Cheap local correctness is calibrated for "was this step wrong?", not for "will removing it break the run?". These AUROC/AUPRC numbers are exploratory discordance diagnostics; the safety conclusions rest on the held-out net effects of Section 9.

Deployed PRMs, uncertainty, and counterfactual signals. Because "PRM scores" in deployment are continuous model outputs rather than dataset ratings, we scored the identical 600 steps with three trained PRM checkpoints (Qwen2.5-Math-PRM-7B, Math-Shepherd, RLHFlow-Llama3.1-8B) and two prompted LLM-judge PRMs, plus step entropy/NLL under teacher forcing and a masked answer-probability drop $I_{\text{mask}} = \log p(y^\star \mid h) - \log p(y^\star \mid h \setminus s_t)$ (Table 13, appendix). The best trained PRM tracks the human rating almost exactly (danger AUROC 0.57 vs. 0.56; step-level, favorable orientation—the run-level, deployment-orientation figure for the raw rating is the below-chance 0.42 above), and the other two trained checkpoints sit at or below chance (0.47, 0.51)—so the construct gap is not an artifact of auditing labels instead of scores. Token-level uncertainty carries no removability information at all. Two signals stand out modestly: cross-PRM disagreement (benefit AUPRC 0.34), supporting uncertainty-based *abstention*, and the counterfactual mask drop—informative but gold-answer-dependent, hence offline-only.

9 From Scores to Policies: Risk–Coverage

If a system triggers rollback on the top-k steps ranked by a signal, does it *certifiably* not lose accuracy, and how many steps does it hurt? A tempting metric—the rate at which control-correct runs come back wrong after deletion—is not causal harm: outcomes are independent samples, so a null deletion on a step with $p^c{=}p^t{=}0.5$ would show 25% "harm" from sampling disagreement alone. We therefore budget the *certified net accuracy change per rollback*: a policy is certified at budget b if the one-sided 95% lower confidence bound (problem-cluster bootstrap) of the mean Δ^{tgt} over its selected steps stays above $-b$. Alongside, we report step-level *tail risk*: the share of selected steps with observed $\hat{\Delta}_i < 0$, minus the exact share expected under the null $p^t{=}p^c$ (Binomial$(4, \hat{p})$ per step). Policies: random; correctness rankings (rating-first, Harmful-first, intersection-first); the three trained PRM scores; a deployable trajectory-state predictor (model D) and its outcome-informed variant (model E; Section 8); an in-sample oracle.

On the *balanced* cohort certification is degenerate: the mix is so error-rich (mean effect +7.2 pp, LCB +4.4) that every policy—including random—certifies deleting *all 600 steps*. It becomes meaningful only in the deployment frame (natural chosen-path prevalence, next paragraph), where deleting all traffic has LCB −4.4 pp and certifies nothing below 5 pp.

Balanced cohort vs. natural prevalence. The cohort is case–control balanced (200 steps per rating); deployment traffic is not. Our reference frame is the annotator-chosen trajectory path of the phase-2 test split—the steps a deployed sampler would actually carry—comprising 16,481 rated steps: 31 rated −1, 1,224 rated 0, 15,226 rated +1 (Table 17, appendix). Chosen paths are error-scarce (annotators route around bad candidates); the rated-but-rejected alternatives form a far larger error pool

Policy	Certifies	Met	Coverage	Net/del. (pp)	Tail exc.
Random	9/20	2/9	0.00	-3.7	0.043
rating-first	20/20	15/20	0.49	$+0.9$	0.044
Harmful-first	12/20	6/12	0.20	-1.8	0.064
PRM (Qwen2.5-Math)	20/20	**9/20**	0.18	$\mathbf{-6.3}$	0.089
PRM (Math-Shep.)	19/20	16/19	0.33	$+6.4$	0.000
PRM (RLHFlow)	14/20	8/14	0.05	$+3.3$	0.038
Predictor (D)	15/20	13/15	0.13	$+20.8$	0.000
Predictor (E)†	20/20	18/20	0.58	$+8.8$	0.010

Table 7: Nested policy evaluation (natural prevalence, 1 pp budget, 20 outer 50/50 problem splits): predictors trained on the calibration half with inner 5-fold cross-fitting, coverage chosen there by scanning k against a *pointwise* cluster LCB (an optimistic screen, not a multiplicity-corrected bound), test scored once. Certifies/Met: nonzero certified coverage / realized test net within budget (even random passes calibration in 9/20); other columns are medians. †E reads the control outcome that defines Δ^{tgt} (outcome-informed upper bound, not deployable). Bold: the non-transfer case.

(26,256 rated completions, 23.2% rated -1), and our rating--1 targets are drawn from that pool and spliced onto the chosen prefix (198/200; all but two of the rating-$0/+1$ targets are chosen-path steps). Reweighting to the chosen-path prevalence therefore assumes spliced alternatives behave like the rare naturally-carried -1 steps, and adjusts rating composition only (the cohort is position-balanced by design; a joint rating$\times$position reweighting gives -0.3 pp, CI $[-5.0, +4.5]$, indistinguishable from rating-only). Under these weights the average deletion effect moves from $+7.2$ pp to -0.5 pp (CI $[-5.1, +4.0]$), with danger and benefit discordance near parity (9.8% vs. 9.3%; Table 16, appendix): in natural traffic, indiscriminate deletion has no upside while the danger does not shrink. Natural-weight numbers are a reference-frame *sensitivity analysis* under this transport assumption; balanced-cohort numbers are diagnostic contrasts; neither is a deployment estimate.

What certifies in natural traffic. Under natural weights with in-sample selection (Table 19, appendix), correctness rankings certify only the error tiers at a 1 pp budget: rating-first covers 2.4% of traffic ($+9.8$ pp per rollback), Harmful-first 3.8% ($+17.0$ pp), and the best trained PRM score 0.02% ($+51$ pp). These are error-correction interventions, not evidence that correctness scores can certify arbitrary rollback or that reasoning compression is impossible—the rating-$+1$ mass (92% of traffic) never certifies below a 5 pp budget.

Honest nested selection. Certification must survive selection. Table 7 repeats the exercise with nested train/calibrate/test selection (protocol in the caption). Three failures emerge. *Instability*: certified coverage swings from 0 to 0.9 across splits for most policies—with $\sim$150 calibration problems, a 95% bound is noisy enough that even *random* certifies in 9 of 20 splits, meeting the budget in only 2. *Checkpoint-dependent transfer*: the trained PRM with the best danger diagnostic (Qwen2.5-Math) certifies in every split yet meets the 1 pp budget in only 9 of 20, realizing $-\mathbf{6.3}$ pp per deletion with the largest tail excess (0.089); Math-Shepherd, by contrast, transfers in 16 of its 19 certifying splits ($+6.4$ pp). Two PRMs trained for the same construct yield opposite safety behavior—a certificate's validity depends on the checkpoint, and correctness quality does not predict which. *Weak payoff*: rating-first meets the budget in 15 of 20 splits but realizes only $+0.9$ pp per deletion at half-cohort coverage (an intersection-first ranking gives identical medians, as the median certified set spans the full rating--1 tier). No deployable signal is a broad certifier: the static predictor D certifies the error tier (13/15 met, $+20.8$ pp) at only 13% coverage, and Math-Shepherd's 16 of 19 at 33% is checkpoint-specific—its danger-diagnostic twin Qwen2.5-Math meets only 9 of 20. Predictor E's 58% coverage comes from reading the control-run outcome that defines Δ^{tgt}—an outcome-informed upper bound, not a deployable certifier. The calibration side is a screen, not a guarantee: k is chosen post hoc against a pointwise bound and the 20 outer splits share problems, so we read the realized test net, not calibration coverage. Which checkpoints certify was not predictable in advance (three trained checkpoints tested).

Rollback: one probe run buys most of the safety. Finally, we simulate the cheapest counterfactual validator: generate *one* probe continuation after a candidate deletion, roll back unless the probe answers correctly, and evaluate on held-out runs. Applied to all 600 steps, this raises the net gain per candidate from $+7.2$ pp (LCB $+4.4$) to $+11.5$ pp (LCB $+9.3$) and cuts the observed harmed-step share from 12.2% (stochastic floor 11.0%) to 3.0%—*below* its 6.8% floor, i.e. no detectable excess harm survives—at a cost of one extra generation per candidate (63% acceptance; a do-nothing control-split confirms the floor is calibrated, 4.7% vs. 3.7% expected). Two caveats keep this honest: the probe is scored by the same gold-answer judge—an oracle relative to deployment, so the simulation upper-bounds a one-probe validator and presupposes one—and the gain is not mere extra sampling, since matched truncations that *keep* the target are inert against their own controls ($\Delta^{\text{plc}}_{\text{own}}$ $+0.5$ pp overall, Table 6).

10 Discussion and Conclusion

Our results do not indict PRMs at their trained task: low ratings flag wrong steps, where deletion *benefit* concentrates. The failure is direction-specific—scores carry almost no information about deletion *danger*, governed by trajectory state and generator recoverability. The oracle–policy gap leaves headroom, but obvious candidates do not close it (Section 8) and informative ones cost extra computation or gold answers. The constructive path is architectural: probe-and-rollback recovers most of the oracle's safety at bounded cost.

Across 600 PRM800K steps, four generator configurations, and 23,000+ regenerated continuations, step correctness and removability dissociate: deletion gains concentrate in low-rated *Harmful* steps; the apparent restart component vanishes under own-control placebos, leaving a gain that travels with the erroneous content; and correctness signals—rating, deployed PRM scores, uncertainty—nominate rollback candidates but alone are unreliable safety certificates, checkpoint-dependent in calibration and unpredictable in transfer. Systems that use PRM scores to trigger rollback should use them to nominate candidates, validate counterfactually, and budget harm explicitly.

Limitations

Generator scope. Primary results use Qwen3-8B without thinking; replications add phi-4, DeepSeek-R1-Distill-Qwen-14B, and a within-checkpoint thinking-mode toggle (300 steps, 3 runs each). The two no-thinking models agree qualitatively but not step-wise (anchor DiD semantics: $+25.4$ vs. $+34.9$ pp; sign agreement 65%); thinking mode on the same checkpoint neutralizes anchors, and the R1 distill replicates the collapse—so anchor-driven gains should not be assumed beyond no-thinking generation. All generators are mid-size open models on mathematical reasoning; we make no claims about frontier models or other domains. The R1-Distill replication could not be run at freeze time (the model was unavailable via the API provider; Section 4) and was completed later on local hardware; phi-4 remains the frozen second generator.

Case–control cohort. Rating-balanced sampling (200 per rating) supports subgroup contrasts, not deployment averages: reweighted to the natural prevalence of PRM800K phase-2 test steps (0.2% rated -1), the mean deletion effect is -0.5 pp and anchor steps are rare. Coverage numbers are fractions of the ranked cohort steps under the stated weighting ("traffic" shares are natural-prevalence-weighted), not deployment volumes.

PRM coverage. The deployed-PRM audit covers three trained checkpoints—Qwen2.5-Math-PRM-7B, the original Math-Shepherd PRM (Wang et al., 2024), and an RLHFlow Llama3.1-8B PRM trained on DeepSeek rollouts—plus two *prompted* LLM-judge PRMs; trained *generative* PRMs (Khalifa et al., 2025) remain uncovered.

Semantic label reliability. Step types come from AI-assisted annotation with human calibration; the two label passes disagree on 27.3% of steps, and the blinded annotation pilot shows only moderate human agreement ($\kappa = 0.40$). The human-only consensus sensitivity analysis (Table 5) reproduces the Harmful-anchor direction ($+25.0$ pp raw; $+22.9$ pp legacy placebo-corrected) on 15 exact-consensus Harmful steps, but this is too small to establish a full-cohort human ground truth. Type-based subgroup boundaries are therefore approximate; the pilot covers 120 of 600 steps, and no adjudicated relabeling of the full cohort exists. The headline interaction survives under both frozen label passes and this human-only sensitivity check, but finer type-level claims inherit label noise.

Judge noise and self-evaluation. The 93.0% agreement and ≤ 0.08 pp substitution sensitivity are measured on 200 audited Qwen3-8B outputs and do not bound error on the unaudited remainder; phi-4 and ProcessBench continuations are scored by the same judge pipeline without a fresh per-generator human audit. The deterministic-grader regrade (Section 4; Appendix A) bounds self-evaluation bias on the primary cohort only—the extension cohorts have no deterministic regrade.

Placebo scope. Matched placebos exist for 511/600 primary steps (position SMD ≤ 0.298); the external cohort's placebo matching uses character length ($\pm 20\%$) rather than tokenizer length, and 96/300 external steps admit no match.

Offline policy evaluation. Risk–coverage and rollback numbers reuse frozen runs; an online system would face distribution shift from its own interventions. The rollback probe is drawn from the same run pool it is evaluated against, which can flatter acceptance quality slightly. In-sample coverage selection flatters every policy; the nested train/calibrate/test protocol of Section 9 is the honest reading. Even there, the 20 outer splits share problems (each appears in roughly half the test halves), so split counts are dependent, and any policy's apparent transfer is itself selected on these splits (a winner's curse); predictor E additionally reads the control-run outcome that defines the target effect, so its coverage is an outcome-informed upper bound, not a deployable certifier. With $K{=}4$ runs per condition, per-step harm attribution is noise-limited: the tail-risk excess over the exact stochastic floor is a conservative signal of truly harmed steps, not a per-step identification. Rollback presupposes a verifier and spends one probe generation per candidate, which must be netted against any claimed efficiency savings in deployment.

Paraphrase validity. C3 paraphrases passed automatic meaning/length checks (7.9% excluded), and a blinded 140-item human fidelity audit rated 97.1% faithful or minimally deviating, with zero meaning changes among the 46 audited anchor-group paraphrases; two neutral-group paraphrases (4.3% of that group's audited items) were judged meaning-changed. C3 conclusions rest on this audited premise.

Ethical Considerations

The study uses the public PRM800K dataset (MIT-licensed problems from MATH) and open-weight models; no personal data is processed. Beyond the authors, four annotators and one adjudicator—unpaid volunteers from the authors' research group—labeled reasoning steps and paraphrase fidelity for public math problems (about one hour each); they were informed of the study's purpose and consented, and the annotation sheets collected no personal information. Counterfactual rollback of reasoning steps is an accuracy/safety intervention; our safety framing (harm budgets) is about answer correctness, not user harm. We release artifacts to support reproduction and auditing.

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A Judge Audit Detail

Audited outputs	200
Binary agreement	93.0%
Correct-class F1	0.931
Pair-transition agreement (60 pairs)	90.0%
Max condition bias	2.3 pp
Δ overall target effect after substitution	-0.08 pp
Δ placebo-corrected effect after substitution	-0.05 pp

Table 8: Judge audit summary. Gates (agreement $\geq$ 90%, bias $\leq$ 5 pp) pass; all headline numbers are insensitive to full substitution of adjudicated labels.

Deterministic grader. Because the LLM judge and the primary generator share a checkpoint, we regrade all 6,314 primary runs with a deterministic grader: PRM800K-style answer normalization followed by a SymPy equivalence check. It agrees with the LLM judge on 93.0% of runs, and substituting its verdicts reproduces the overall and anchor deletion effects within 0.9 pp, bounding self-evaluation bias on the primary cohort; the full comparison ships with the audit artifacts.

B Robustness and Disclosure Tables

Contrast (pp)	Anchor	Stable-corr.	Fragile-neutral
C1−C0 (raw deletion)	$+20.3$	-11.9	$+24.1$
C1−C2 (vs. placebo)	$+15.6$	$+16.8$	$\mathbf{+0.0}$
C3−C0 (paraphrase)	$\mathbf{+1.7}$	-8.3	$+20.9$
C1−C3 (vs. paraphrase)	$\mathbf{+19.2}$	-2.7	$+4.1$
C2−C_p (placebo, own ctrl.)	-0.7	-0.8	$+3.9$

Table 9: Four-condition strong controls (240 steps $\times$ 4 runs; 80 per group; paraphrase available for 221). Anchors (selected on labels, not outcomes): paraphrase $\approx$ control while deletion beats both—the *content* is the anchor. Fragile-neutral and Stably-correct are selected on their C0 draws, so their vs-C0 rows are biased by regression to the mean (upward and downward, respectively); the own-control row shows the operator itself is inert. Bold marks contrasts discussed in the text.

Annotation pilot details. The blinded pilot re-annotated 120 cohort steps: two independent human labels per step from four annotators in two pairs, 61.7% raw agreement ($\kappa = 0.40$), with disagreement concentrated on the essential/redundant boundary. The main-text Table 5 reports the exact-consensus sensitivity analysis; 93% of consensus-Harmful steps carry rating -1. An independent adjudicator resolved all 46 disagreements, yielding final labels for all 120 pilot steps, but those adjudications were not used to relabel the full 600-step cohort.

Position	Group	n	Δ^{tgt} (pp)
Early	Anchor	56	$+16.1$ [$+4.9$, $+28.1$]
Early	Other	145	$+0.7$ [-6.4, $+7.9$]
Middle	Anchor	63	$+24.6$ [$+13.9$, $+35.3$]
Middle	Other	138	$+2.5$ [-4.3, $+9.8$]
Late	Anchor	59	$+28.8$ [$+16.5$, $+41.5$]
Late	Other	139	-2.0 [-7.0, $+2.9$]
Early–Middle	Interaction difference		-6.7 [-25.3, $+11.5$]
Early–Late	Interaction difference		-15.4 [-33.6, $+3.8$]
Middle–Late	Interaction difference		-8.7 [-26.9, $+10.1$]

Table 10: Anchor-by-position robustness using the primary initial labels. Anchor = rating $-1\times$Harmful; Other = all remaining steps. Effects are target deletion minus control, with 95% percentile cluster-bootstrap intervals over steps (5,000 replicates, seed 20260801). Interaction differences compare the Anchor–Other contrast between the named positions; every interval includes zero. Position bins are the pre-existing balanced early/middle/late bins, not redefined for this analysis.

Predictor	Coefficient (pp)	95% CI (pp)	Reference
Rating -1	$+17.9$	[$+5.3$, $+30.9$]	vs. rating $+1$
Rating 0	-0.8	[-8.7, $+7.5$]	vs. rating $+1$
Redundant	$+3.8$	[-5.1, $+12.7$]	vs. Essential
Harmful	$+7.1$	[-4.1, $+18.4$]	vs. Essential
Middle	$+3.9$	[-4.5, $+12.2$]	vs. Early
Late	$+2.2$	[-5.6, $+10.3$]	vs. Early
log(1+ target tokens)	$+1.2$	[-4.9, $+7.5$]	per log unit

Table 11: Length-adjusted deletion-effect regression over the 600-step primary cohort. The outcome is $100\Delta_i^{\text{tgt}}$; predictors are rating indicators, initial AI-assisted semantic-type indicators, position indicators, and log target-token count. Estimates are OLS coefficients with 5,000 problem-cluster bootstrap replicates (303 problem clusters, seed 20260802). The length coefficient is small and its interval includes zero; the Harmful coefficient is also not distinguishable from zero after adjustment.

Predictor features. Model D uses 12 static features: indicators rating$=0$ and rating$=+1$; indicators type=Redundant and type=Harmful (initial pass); their four interactions; position-bin indicators (middle, late); $\log(1 +$ target tokens$)$; $\log(1 +$

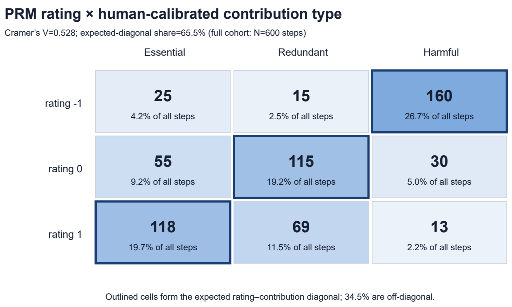

Figure 3: PRM800K rating vs. human-calibrated semantic contribution for the 600-step cohort. The association is strong (Cramér's $V = 0.528$) but 34.5% of steps are off-diagonal: local correctness does not determine contribution.

Model (features)	Danger AUPRC	Benefit AUPRC
Base rate	0.151	0.360
A: rating only	0.139	0.432
B: type only	0.208	0.407
C: rating + type	0.140	0.380
D: static context	0.160	0.412
E: + trajectory state	**0.263**	**0.478**

Table 12: Out-of-fold prediction of run-level danger and benefit discordance, conditioned on the control draw: danger = P(deletion draw wrong | control draw correct), base rate 15.1%; benefit base rate 36.0%. The rating is *below chance* for danger (AUROC 0.42, CI $[0.36, 0.49]$); only trajectory-state features (model E, which reads the control-run outcome and so requires extra runs) achieve a meaningful margin (danger AUPRC +0.10 over D).

prefix tokens). Model E adds the control-correct frequency (13 features). Both are L2-regularized logistic models trained with 5-fold cross-validation grouped by problem; every policy score is an out-of-fold prediction. Model E's added feature is the control-run success frequency, one of the two terms in $\Delta^{\text{tgt}} = \bar{Y}^t - \bar{Y}^c$; problem-level cross-fitting cannot remove this within-step dependence on the evaluated outcome, so E is reported as an outcome-informed (oracle-style) upper bound rather than a deployable predictor. Model D uses no outcome-derived feature.

Judge configuration. The judge is Qwen3-8B served at temperature 0, prompted with the problem, the gold answer, and the generated final answer, and instructed to return a JSON verdict on mathematical equivalence; outputs with no extractable final answer are failed by rule. The audit sampled 200 outputs stratified by condition and verdict; two annotators labeled blind to condition, with disagreements adjudicated. Full prompts, seeds, and adjudication records ship with the release.

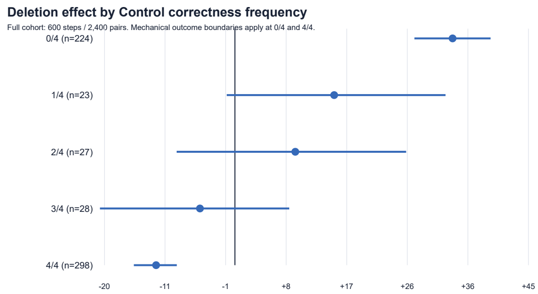

Figure 4: Deletion effect by control-run stability. Gains exist only where control already fails. Outcome-conditioned strata: conditioning on the control draws biases the extreme strata outward (regression to the mean), so these are descriptive bounds, not causal subgroup effects.

	Danger		Benefit	
Signal (step-level)	AUROC	AUPRC	AUROC	AUPRC
Base rate	—	0.15	—	0.24
PRM800K rating	0.56	0.17	0.61	0.34
Qwen2.5-Math-PRM-7B	0.57	0.18	0.61	**0.37**
Math-Shepherd PRM (7B)	0.47	0.14	0.54	0.26
RLHFlow PRM (8B)	0.51	0.15	0.59	0.28
LLM-judge PRM (Gemini)	0.53	0.16	0.51	0.25
LLM-judge PRM (phi-4)	0.49	0.14	0.48	0.23
PRM disagreement (5 models)	0.53	0.16	**0.62**	0.34
Step entropy	0.46	0.15	0.48	0.24
Step NLL	0.50	0.15	0.49	0.22
Masked answer-prob drop	0.56	0.19	0.57	0.30

Table 13: Step-level signals vs. deletion outcomes on the identical 600-step cohort (step-level danger/benefit = any control-correct/deletion-wrong, resp. control-wrong/deletion-correct, discordance among the step's pairs; each signal scored in its favorable orientation, so higher warns against / flags for deletion). The best benefit AUPRCs reach 0.34–0.37 over a 0.24 base—a real lift—while the best danger AUPRC (0.19) barely exceeds its 0.15 base.

C Reproducibility

Primary numbers derive from two frozen master tables (600 steps; 6,314 runs) via a dependency-free Python pipeline with fixed seeds (cluster bootstrap seed 20260722, downstream 20260723); a single command regenerates every table, figure source, and the numeric manifest consumed by this paper. The complete accounting is shown in Table 21: 6,314 primary runs plus 17,126 extension runs equals 23,440 generation slots in total. The extension cohorts add 17,126 runs (cross-generator 2,691; long-CoT replication 2,691; thinking-mode toggle 3,582 including its 891 own-controls; strong

Estimate	Cluster	n	Effect (pp)	95% CI
Anchor $\Delta^{\rm tgt}$, initial	step	178	$+23.31$	$[+16.6, +29.9]$
Anchor $\Delta^{\rm tgt}$, initial	problem	178	$+23.31$	$[+16.2, +30.5]$
Anchor $\Delta^{\rm sem}$ (legacy), initial	step	151	$+13.85$	$[+7.0, +20.8]$
Anchor $\Delta^{\rm tgt}$, calibrated	step	160	$+20.94$	$[+13.8, +28.1]$
Anchor $\Delta^{\rm tgt}$, calibrated	problem	160	$+20.94$	$[+13.5, +28.4]$
Anchor $\Delta^{\rm sem}$ (legacy), calibrated	step	137	$+14.84$	$[+7.5, +22.1]$
Overall $\Delta^{\rm tgt}$	step	600	$+7.21$	$[+3.9, +10.5]$
Overall $\Delta^{\rm tgt}$	problem	600	$+7.21$	$[+3.7, +10.7]$

Table 14: Label-set and clustering robustness (independent 5,000-replicate bootstrap draws, so intervals differ slightly from Table 3; point estimates identical). The anchor effect survives the calibrated label pass and problem-level clustering. Rows marked *legacy* use the shared-control placebo contrast of Eq. 1, superseded by the own-control difference-in-differences of Table 6 ($+25.4$ pp); they are retained for comparability with the frozen analysis plan.

Rating of placebo-cutoff step	Runs	Share
$+1$	900	59.4%
0	167	11.0%
unrated	447	29.5%
-1	0	0.0%

Table 15: PRM800K ratings at the cutoffs of the 1,514 placebo runs: the placebo never truncates at a known-bad step.

controls 1,843 = 884 paraphrase C3 + 959 placebo C2, one of the 960 scheduled C2 generations having failed at the provider; ProcessBench 2,412; placebo own-controls 3,907 across four cohorts), each with its own checkpointed generation, judging, and analysis script under the corresponding workstream directory. Robustness artifacts (label-set re-estimation, placebo-cutoff composition, natural-prevalence reweighting, and the calibration/test policy split) are regenerated by dedicated scripts in the same release. The release includes the intervention builder, matched-placebo selector, judge-audit tooling, policy risk–coverage and rollback code, the 120-step blinded annotation pilot, PRM-scoring adapters, the signal-extraction script, and the StepRem benchmark split with its evaluation script.

	Balanced cohort	Natural prevalence
Overall $\Delta^{\rm tgt}$ (pp)	$+7.21$ $[+3.9, +10.7]$	-0.52 $[-5.1, +4.0]$
Run danger rate	8.5%	9.8%
Run benefit rate	15.7%	9.3%
Dangerous-step share	15.2%	17.5%
Beneficial-step share	23.5%	17.6%

Table 16: Balanced-cohort vs. natural-prevalence estimates (weights $0.2\%/7.4\%/92.4\%$ for ratings $-1/0/+1$, from 16,481 rated phase-2 test steps). Run danger/benefit rates are *joint* discordance rates over all pairs (Section 3); the conditional rates used for prediction (15.1%/36.0%) divide these by the control accuracy (danger) or inaccuracy (benefit). CIs are an independent bootstrap draw.

Frame	n_{-1}	n_0	n_{+1}	Total
Phase-2 test, chosen path	31	1,224	15,226	16,481
Phase-2 test, all completions	6,080	2,002	18,174	26,256
Cohort targets from chosen path	2	198	200	400
Cohort targets from alternatives	198	2	0	200

Table 17: Sampling-frame audit (integer counts). The 600 cohort targets are unique steps over 303 problems; every cohort problem is in the phase-2 test split. Rating--1 targets come almost entirely from rated-but-rejected alternative completions spliced onto the chosen prefix; reweighting (Table 16) uses the chosen-path frame.

Cohort / check	$\Delta^{\rm plc}_{\rm own}$ (pp)	$\Delta^{\rm sem}_{\rm DiD}$ anch. (pp)
Primary, anchors (151)	-1.6 $[-6.5, +3.2]$	$+25.4$ $[+16.1, +34.9]$
phi-4, anchors (100)	-1.3 $[-6.6, +3.8]$	$+34.9$ $[+25.8, +44.7]$
Strong-control C2, all groups (240)	$+0.8$ $[-2.7, +4.4]$	—
Anchor cutoffs before / after	-5.2 / $+4.3$	—
Batch drift (128 identical prompts)		new $-$ old $= +3.1$ pp (n.s.)

Table 18: Own-control placebo (DiD) robustness. Matched innocuous deletions are inert across the no-thinking PRM800K cohorts shown here; the long-CoT generator instead shows a genuine content-agnostic truncation *benefit* ($+8.0$ pp, Section 5) and the external first-error subset a truncation *penalty* (-8.4 pp, Section 7). The drift row compares fresh C_p runs against frozen runs with byte-identical prompts; all C_p runs are a later batch, so any positive drift lowers $\Delta^{\rm plc}_{\rm own}$ and raises every DiD one-sidedly—a uniform $+3.1$ pp adjustment would move the overall DiD $+7.5 \to +4.4$ and the anchor DiD $+25.4 \to +22.3$, changing no conclusion.

Policy	Budget 1 pp		Budget 3 pp	
	Traffic	Net/del. (LCB), pp	Traffic	Net/del. (LCB), pp
Random	0	—	0	—
rating-first	2.4%	$+9.8\,(+2.0)$	12.2%	$+1.4\,(-1.9)$
Harmful-first	3.8%	$+17.0\,(+1.2)$	3.8%	$+17.0\,(+1.2)$
PRM (Qwen2.5-Math)	0.02%	$+51.3\,(+35.5)$	16.3%	$+8.2\,(-1.8)$
PRM (Math-Shep.)	21.8%	$+9.6\,(+3.0)$	31.3%	$+5.2\,(-2.8)$
PRM (RLHFlow)	40.4%	$+5.0\,(-0.6)$	47.7%	$+3.1\,(-2.4)$
Predictor (D)	66.1%	$+4.2\,(-0.8)$	88.9%	$+1.1\,(-2.9)$
Predictor (E)	68.5%	$+4.5\,(-0.1)$	79.7%	$+1.5\,(-2.7)$
Oracle	96.4%	$+3.2\,(-0.2)$	96.4%	$+3.2\,(-0.2)$

Table 19: In-sample causal certification under natural prevalence: largest traffic coverage whose one-sided 95% cluster LCB of net effect per deletion clears the budget, with the realized net (LCB in parentheses). In-sample k-selection upper-bounds every policy; the nested evaluation (Table 7) is the honest reading.

	n	Anchor	Stable	Neutral	Rating -1
Retained	221	33.0%	33.9%	33.0%	34.8%
Excluded	19	36.8%	26.3%	36.8%	36.8%

Table 20: Composition of retained vs. automatically excluded C3 paraphrases: share of each set in the three strong-control groups and in the rating--1 tier (which overlaps the Anchor column); mean target length 38.0 vs. 32.7 tokens.

Experiment component	Steps	Conditions	Generation runs
Primary deletion (Qwen3-8B, no-think)	600	C_0, C_1: 4 each	4,800
Primary matched placebo deletion (C_2)	511	1–4 matched placebo runs per step	1,514
Cross-generator deletion (phi-4)	300	C_0, C_1: 3 each	1,800
Cross-generator matched placebo (C_2; phi-4)	297	3 matched placebo runs per step	891
Long-CoT deletion (R1-Distill)	300	C_0, C_1: 3 each	1,800
Long-CoT matched placebo (C_2; R1-Distill)	297	3 matched placebo runs per step	891
Thinking-mode toggle core (Qwen3-8B)	300	M2 task set; archived thinking-mode slots	2,691
Thinking-mode own controls (C_p)	297	3 own-control runs per matched cutoff	891
Strong controls (placebo and paraphrase)	240	New C_2: 959; new C_3: 884 completed runs	1,843
ProcessBench deletion	300	C_0, C_1: 3 each	1,800
ProcessBench matched placebo (C_2)	204	3 matched placebo runs per step	612
M12 own-control C_p (primary)	511	One own control per matched placebo cutoff	1,513
M12 own-control C_p (phi-4)	297	One own control per matched placebo cutoff	891
M12 own-control C_p (R1-Distill)	297	One own control per matched placebo cutoff	891
M12 own-control C_p (ProcessBench)	204	One own control per matched placebo cutoff	612
Total (no double counting)	—	—	**23,440**

Table 21: Experiment accounting with no double counting. “Generation runs” counts archived generation slots used by the frozen analyses. C_2 denotes matched placebo deletion; C_p denotes its additional own control and is listed separately. Reused control/target runs are counted only in their original component. The primary cohort is $4{,}800$ deletion runs plus $1{,}514$ matched-placebo runs; phi-4 and R1 each have $1{,}800$ deletion runs plus 891 matched-placebo runs; the thinking toggle has $2{,}691$ core slots plus 891 own controls; and ProcessBench has $1{,}800$ deletion runs plus 612 matched-placebo runs. The strong-control row reports the $959 + 884$ completed new runs after provider/paraphrase attrition.